Unit no. ° 15

CONSOLIDATION AND TRANSITION TO PREMIÈRE

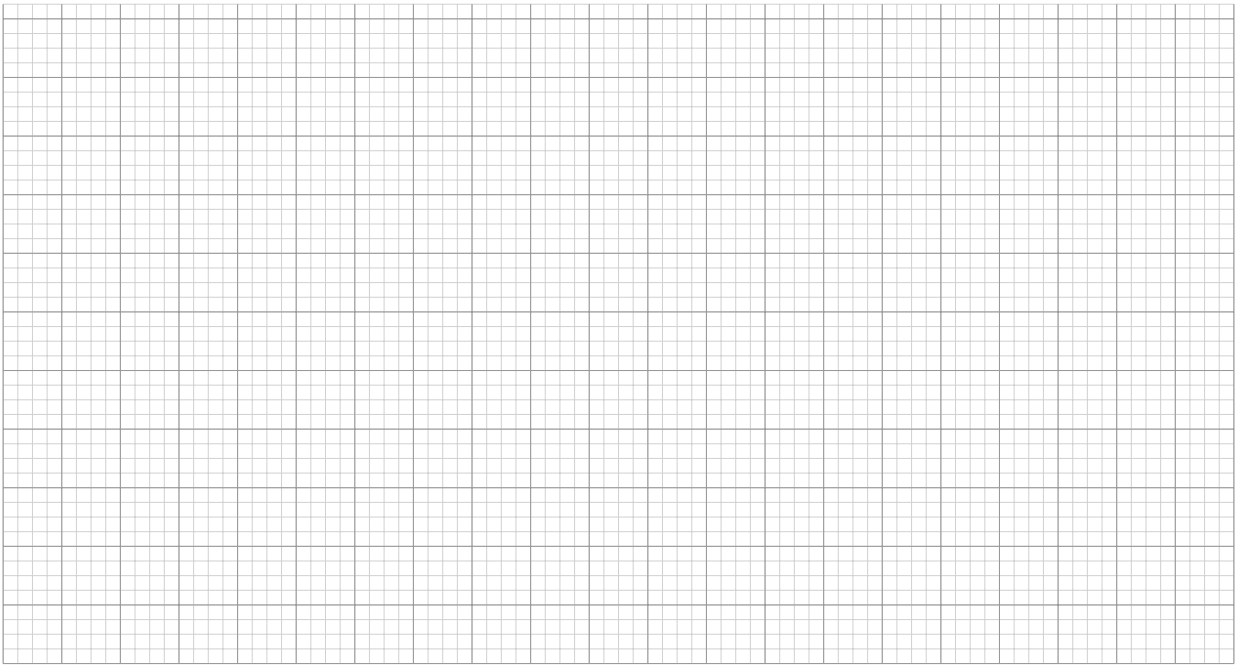
25 May to 11 June 2027

I Review of calculation, functions and geometry

Example 15.1: Annual Review · Part A

1. Write 168 /252 in lowest terms; calculate 2 $^{-3} + 1/4$.
2. Factorise $x^2 - 6x$ + 9, then solve ($x - 3)^2 = 4$.
3. Solve $\dfrac{x-2}{x+1} \geq 0$.
4. For $f(x) = 4 - (x-1)^2$ on [−2; 3], give the maximum and the x-values where $f(x) = 0$.
5. With $A(0;1),\ B(4;3),\ C$(6; 4), prove that the points are collinear and give an equation of the line.

Solution:

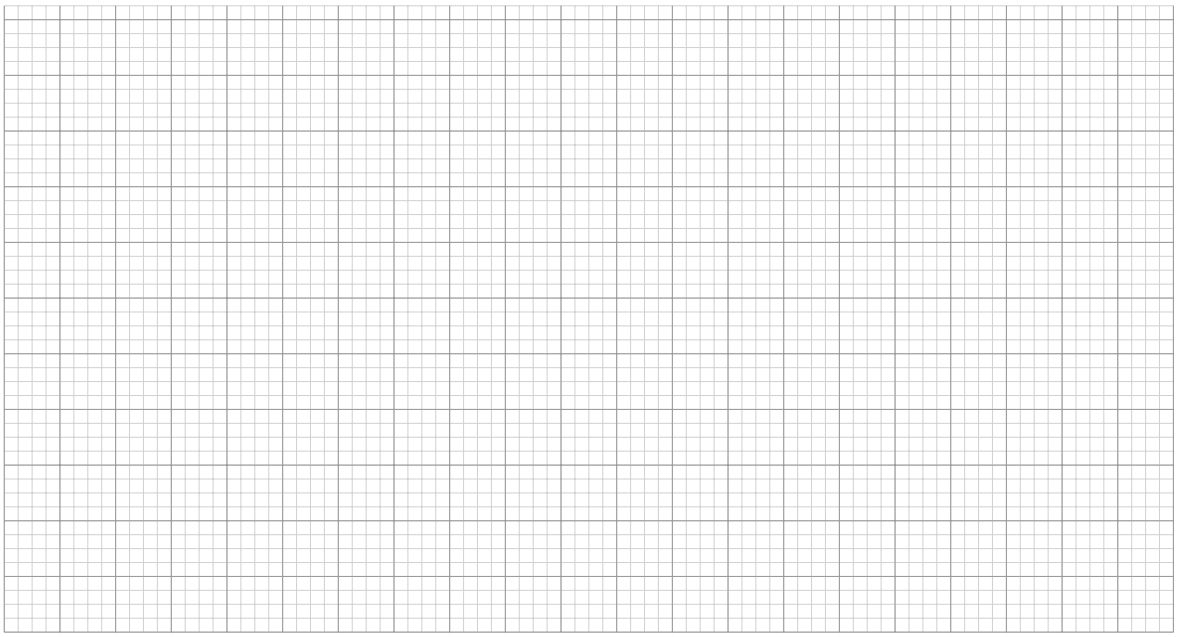


II Review of data, probability and modelling

Example 15.2: Annual Review · Part B

1. Calculate the mean and median of 2; 4; 4; 6; 9. Determine Q_1, Q_3.
2. A price falls by 15%, then rises by 20%. Give the overall rate and the rate that would then be needed to restore the price.
3. In a fictional population of 100 people, 40 do sport (S), 30 come by bike (V), 18 do both. Build the two-way table and calculate $P_S(V)$, $P_V(S)$ for an equally likely draw.
4. For a rectangle with perimeter 24 m, express the area in terms of one dimension x and prove its maximum.

Solution:



III Choosing a strategy and preparing what's next

Example 15.3: A program to explain

For a given function f, explain the following program. What does it return for $f(x) = x^2$?

```
def tableau(f):
    valeurs = []
    for x in range(-3, 4):
        valeurs.append((x, f(x)))
    return valeurs
```

Solution:

Six skills to use

Investigate : try out and organise approaches. **Model** : choose variables, domain and assumptions. **Represent** : move from the text to a table, a figure or a graph. **Reason** : justify and distinguish conjecture from proof. **Calculate** : choose a form and check the operations. **Communicate** : explain clearly with appropriate notations and units.

Example 15.4: End-of-year portfolio

Choose a successful piece of work and a corrected piece of work. For each, identify the skills used, explain a method and set a personal objective for the following year. Then redo an exercise corresponding to this objective without looking at the solutions.

Solution: